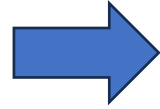


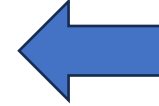
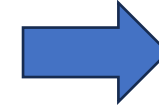
Objective of this exercise



Write code
Make changes
Check the code in



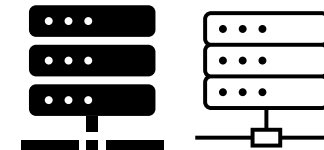
Detects the change
Raises a build event



Build and runs tests



CodePipeline

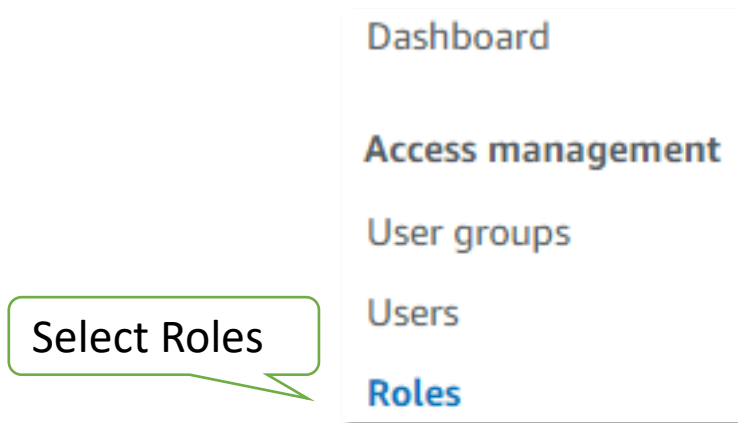
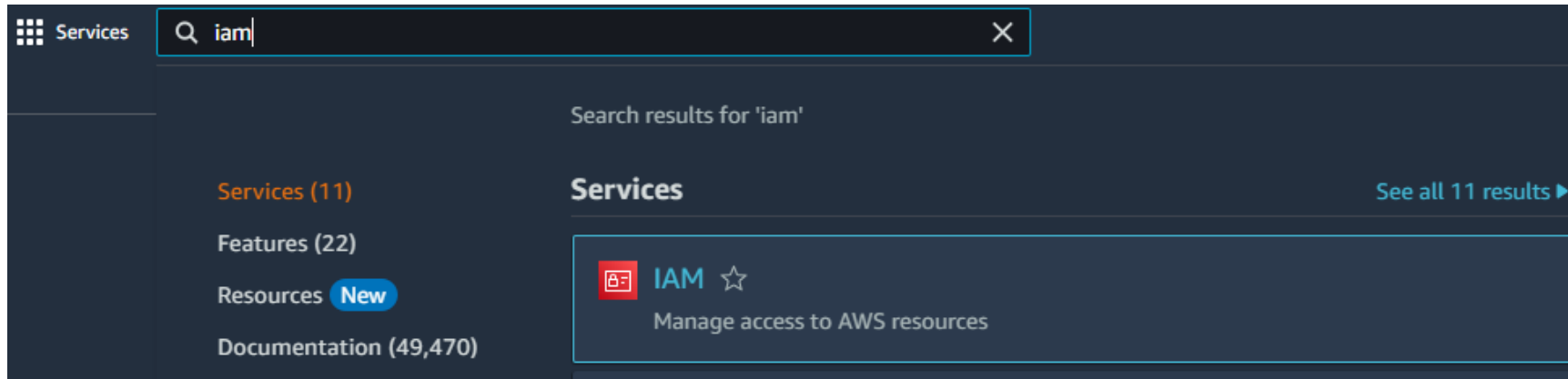


Production

In this lab, you will

- Deploy a web app from GitHub to CodePipeline
- Build and test the app
- Use EBS to deploy the app
- Make changes to your code and commit the changes to GitHub
- Set up the CodePipeline to detect the change and rebuild the app for EBS to deploy.

Create a Service role for your Elastic Bean Stalk service



☒ **AWS service**

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **SAML 2.0 federation**

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Choose a use case for the specified service.

Use case

☒ **EC2**

Allows EC2 instances to call AWS services on your behalf.

☐ **EC2 Role for AWS Systems Manager**

For the Permissions for the role add:

Q Bean		
− Policy name ↗		
☐	+	AdministratorAccess-AWSElasticBeanstalk
☐	+	AWSElasticBeanstalkCustomPlatformforEC2Role
☒	+	AWSElasticBeanstalkEnhancedHealth
☒	+	AWSElasticBeanstalkManagedUpdatesCustomerRolePolicy

Type a name for the new Role and then click the **Create role** button.

Role details

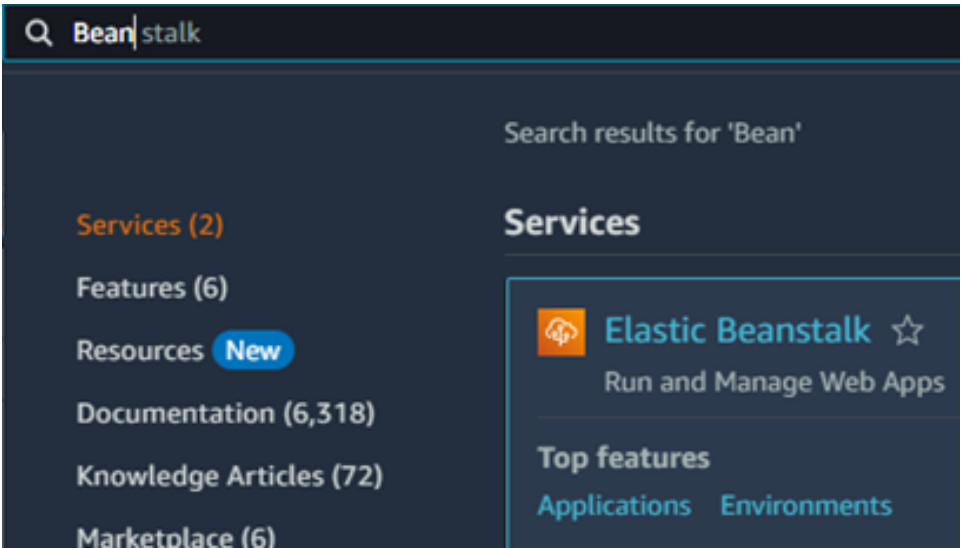
Role name

Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+'

Create role

Create your BeanStalk application to upload your app



Type a name for your app (any name you like)

Application name

Get started

Easily deploy your web application in minutes.

Create application

Select Node.js for your platform for a sample app and then click the **Next** button

Platform [Info](#)

Platform type

☒ Managed platform

Platforms published and maintained by Amazon Elastic Beanstalk. [Learn more](#)

☐ Custom platform

Platforms created and owned by you. This option is unavailable if you have no platforms.

Platform

Node.js

Platform branch

Node.js 20 running on 64bit Amazon Linux 2023

Platform version

6.1.2 (Recommended)

Application code [Info](#)

☒ Sample application

☐ Existing version

Application versions that you have uploaded.

☐ Upload your code

Upload a source bundle from your computer or copy one from Amazon S3.

Skip to review

Cancel

Previous

Submit

✓ Environment successfully launched.

[Elastic Beanstalk](#) > [Environments](#) > EBS1-env

EBS1-env [Info](#)

Environment overview

Health

✓ Ok

Domain

EBS1-env.eba-iuwg7zyp.us-east-1.elasticbeanstalk.com [↗](#)

Click

[Events](#)

[Health](#)

[Logs](#)

[Monitoring](#)

[Alarms](#)

[Managed updates](#)

Events (11) [Info](#)

🔍 *Filter events by text, property or value*

Deploy the sample NodeJS app using the AWS Elastic Beanstalk

Congratulations

Your first AWS Elastic Beanstalk Node.js application is now running on your own dedicated environment in the AWS Cloud

This environment is launched with Elastic Beanstalk Node.js Platform

What's Next?

- [AWS Elastic Beanstalk overview](#)
- [AWS Elastic Beanstalk concepts](#)
- [Deploy an Express Application to AWS Elastic Beanstalk](#)
- [Deploy an Express Application with Amazon ElastiCache to AWS Elastic Beanstalk](#)
- [Deploy a Geddy Application with Amazon ElastiCache to AWS Elastic Beanstalk](#)
- [Customizing and Configuring a Node.js Container](#)
- [Working with Logs](#)

Then start



CodePipeline ☆

Release Software using Continuous Delivery



Pipeline settings

Pipeline name

Enter the pipeline name. You cannot edit the pipeline name after it is created.

No more than 100 characters

Service role



New service role

Create a service role in your account



Existing service role

Choose an existing service role from your account

Role name

Type your service role name



Allow AWS CodePipeline to create a service role so it can be used with this new pipeline

Add source stage [Info](#)

Source

Source provider

This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (Version 2) ▼



New GitHub version 2 (app-based) action

To add a GitHub version 2 action in CodePipeline, you create a connection, which uses GitHub Apps to access your repository. Use the options below to choose an existing connection or create a new one. [Learn more](#)

Connection

Choose an existing connection that you have already configured, or create a new one and then return to this task.

🔍 ✕

or

[Connect to GitHub](#)



Ready to connect

Your GitHub connection is ready for use.

Repository name

Choose a repository in your GitHub account.

🔍 ✕

<account>/<repository-name>

Branch name

Choose a branch of the repository.

🔍 ✕

Add deploy stage [Info](#)



You cannot skip this stage

Pipelines must have at least two stages. Your second stage must be either a build or deployment stage. Choose a provider for either the build stage or deployment stage.

Deploy

Deploy provider

Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

▲

Q

AWS AppConfig

AWS CloudFormation

AWS CloudFormation Stack Set

AWS CodeDeploy

AWS Elastic Beanstalk

AWS OpsWorks Stacks

AWS S3PublishDev

AWS Service Catalog

Amazon ECS

Amazon ECS (Blue/Green)

Amazon S3

Previous

Next

Add build stage [Info](#)

Build - *optional*

Build provider

This is the tool of your build project. Provide build artifact details like operating system, build spec file, and output file names.

Cancel

Previous

Skip build stage

Next

Skip build stage



Your pipeline will not include a build stage. Are you sure you want to skip this stage?

Cancel

Skip

Add deploy stage [Info](#)



You cannot skip this stage

Pipelines must have at least two stages. Your second stage must be either a build or deployment stage. Choose a provider for either the build stage or deployment stage.

Deploy

Deploy provider

Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

AWS Elastic Beanstalk



Region

US West (N. California)



Application name

Choose an application that you have already created in the AWS Elastic Beanstalk console. Or create an application in the AWS Elastic Beanstalk console and then return to this task.

Q EBS_GIT



Environment name

Choose an environment that you have already created in the AWS Elastic Beanstalk console. Or create an environment in the AWS Elastic Beanstalk console and then return to this task.

Q EBSGIT-env



Cancel

Previous

Next

Step 4: Add deploy stage

Deploy action provider

Deploy action provider

AWS Elastic Beanstalk

ApplicationName

EBS_GIT

EnvironmentName

EBSGIT-env

Cancel

Previous

Create pipeline



Success

Congratulations! The pipeline EBS_GIT_pipe has been created.

Developer Tools > CodePipeline > Pipelines > EBS_GIT_pipe

EBS_GIT_pipe

Source

In progress

Pipeline execution ID: 0166d12b-8331-4074-a22f-3eabcca54c86

Source



In progress - Just now

Disable transition

Deploy

Didn't Run

Deploy

[AWS Elastic Beanstalk](#)

Didn't Run

No executions yet

After a few minutes
click here and click on
your EBS URL

← → ↺ ⚠ Not secure | ebsgit-env.eba-vitd4gb3.us-west-1.elasticbeanstalk.com/quiz

```
[
  {
    id: "demo-quiz-1",
    name: "Demo Quiz #1"
  },
  {
    id: "demo-quiz-2",
    name: "Demo Quiz #2"
  }
]
```



```
3  const cors = require('cors');
4  const quiz = require('./services/quiz')
5
6  // This allows us to read environment v
7  // (But, when the system runs in the cl
8  const dotenv = require('dotenv');
9  dotenv.config();
10
11 const app = express();
12 app.use(express.json());
13 app.use(cors());
14
15 // This is just for SSL certificate ver
16 app.use('/.well-known/pki-validation',
17
18 // Create or update a quiz, ready for p
19 app.post('/quiz', async (req, res) => {
20   const id = await quiz.saveQuizAsync
21   res.status(200).json({ id });
22 });
23 app.get('/qa', async (req, res) => {
24   res.status(400).json({
25     message: 'Welcome :)'
26   });
27 });
28
```

Make changes in GitHub



⚠ Not secure | ebsgit-env.eba-vitd4gb3.us-west-1.elasticbeanstalk.com/qa

```
{
  message: "Welcome :)"
}
```